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Secondary 6 - Chemistry

Atoms, Ions and Bonding

Name: ______________________________

Class: ______________________________

1. Element M has two isotopes, $^{71}\mathrm{M}$ and $^{73}\mathrm{M}$, with abundances of 40.0% and 60.0%, respectively. What is the relative atomic mass of M, to one decimal place?

A. 71.0

B. 71.8

C. 72.0

D. 72.2

2. For the ion $^{39}_{19}\mathrm{K}^{+}$, which combination gives the numbers of protons, neutrons and electrons, together with its electronic arrangement?

A. 19 protons, 20 neutrons, 18 electrons; electronic arrangement $2, 8, 8$

B. 19 protons, 39 neutrons, 18 electrons; electronic arrangement $2, 8, 8$

C. 19 protons, 20 neutrons, 20 electrons; electronic arrangement $2, 8, 8$

D. 18 protons, 20 neutrons, 19 electrons; electronic arrangement $2, 8, 9$

3. An ionic compound is formed from aluminium ions, Al^{3+}, and sulphate ions, SO_4^{2-}. Which combination gives the correct formula and name?

A. AlSO_4, aluminium sulphate

B. $\mathrm{Al}_2(\mathrm{SO}_4)_3$, aluminium sulphate

C. $\mathrm{Al}_3(\mathrm{SO}_4)_2$, aluminium sulphate

D. $\mathrm{Al}_2\mathrm{SO}_4$, aluminium sulphide

4. Which sample is expected to conduct electricity?

A. Solid sodium chloride

B. Molten sodium chloride

C. Solid potassium nitrate

D. Pure liquid ethanol

5. A water molecule reacts with a proton to form a hydronium ion, H_3O^+. In the newly formed O–H dative covalent bond, which statement is correct?

A. Both shared electrons come from the oxygen atom.

B. One shared electron comes from oxygen and one comes from the proton.

C. Both shared electrons come from the proton.

D. No electrons are shared; the bond is purely ionic.

6. Using $A_r(\mathrm{C}) = 12.0$, $A_r(\mathrm{H}) = 1.0$ and $A_r(\mathrm{O}) = 16.0$, calculate the relative molecular mass of ethanoic acid, $\mathrm{CH_3COOH}$.

A. 58.0

B. 60.0

C. 61.0

D. 64.0

7. Which statement best explains why a solid copper wire conducts electrical current?

A. The positive copper ions move through a stationary sea of electrons.

B. Delocalised electrons move through a lattice of positive copper ions.

C. Neutral copper molecules move from one end of the wire to the other.

D. Localised electrons in covalent bonds move while the copper ions stay neutral.

8. Which statement correctly describes the simple model of metallic bonding in a solid metal?

A. Neutral atoms are held together by shared pairs of electrons.

B. Positive metal ions form a lattice and are electrostatically attracted to a sea of delocalised electrons.

C. Negative ions form a lattice and attract fixed positive electrons.

D. Positive metal ions are held together by electrostatic repulsion.

9. A substance is solid at room temperature, has a melting point of $78\,^{\circ}\mathrm{C}$, dissolves in water, and its aqueous solution does not conduct electricity. Which structure is most likely?

A. Giant ionic structure

B. Giant metallic structure

C. Giant covalent structure

D. Simple molecular structure

10. A solid material has a high melting point, is insoluble in water, conducts electricity in the solid state, and can be drawn into wire. Which structure is most likely?

A. Giant ionic structure

B. Giant metallic structure

C. Giant covalent structure

D. Simple molecular structure

Atoms, Ions and Bonding

Answers

1 **D**

Relative atomic mass is a weighted mean, so each isotope mass is multiplied by its percentage abundance.

$$A_r(\mathrm{M}) = \frac{(71)(40.0) + (73)(60.0)}{100}$$

$$= \frac{2840 + 4380}{100} = \frac{7220}{100} = 72.2$$

Therefore, the relative atomic mass is 72.2.

2 **A**

Atomic number gives the number of protons, while mass number equals protons plus neutrons. A positive ion has lost electrons.

Protons: $p = 19$

Neutrons: $n = 39 - 19 = 20$

Since the ion has a $+1$ charge, it has lost one electron: $e = 19 - 1 = 18$.

Eighteen electrons fill the shells as $2, 8, 8$. Therefore, the correct combination is 19 protons, 20 neutrons, 18 electrons and electronic arrangement $2, 8, 8$.

3 **B**

The total charge of an ionic compound must be zero. The smallest common total charge for $+3$ and -2 is 6.

Two aluminium ions give $2 \times (+3) = +6$.

Three sulphate ions give $3 \times (-2) = -6$.

Therefore, the formula is $\mathrm{Al_2(SO_4)_3}$, named aluminium sulphate. Brackets are needed because the polyatomic sulphate ion occurs more than once.

4 **B**

In a giant ionic structure, the ions cannot move freely while the substance is solid. When sodium chloride is molten, its $\mathrm{Na^+}$ and $\mathrm{Cl^-}$ ions become mobile and carry charge.

Therefore, molten sodium chloride conducts electricity.

5 **A**

A dative covalent bond is a shared pair of electrons in which both electrons are supplied by the same atom.
The oxygen atom in water has a lone pair available for donation. The proton, H^+, has no electron to contribute.
Therefore, both electrons in the newly formed shared pair originally come from the oxygen atom.

6 **B**

First count the atoms in CH_3COOH: there are two carbon atoms, four hydrogen atoms and two oxygen atoms.
$M_r(CH_3COOH) = 2(12.0) + 4(1.0) + 2(16.0)$
$= 24.0 + 4.0 + 32.0 = 60.0$
Therefore, the relative molecular mass is 60.0.

7 **B**

In the simple model of a metal, positive copper ions form a lattice and delocalised electrons can move through the lattice.
When a potential difference is applied, these mobile electrons move and transfer electrical charge. The positive copper ions remain in their lattice positions.

8 **B**

The simple model represents a metal as a lattice of positive metal ions surrounded by a sea of delocalised electrons.
The electrostatic attraction between the positive ions and the delocalised electrons holds the metallic structure together.

9 **D**

A simple molecular structure consists of separate molecules held together by relatively weak intermolecular forces, so it commonly has a comparatively low melting point.
The solution does not conduct electricity because the molecules do not produce mobile ions when they dissolve in water. These properties indicate a simple molecular structure.

10 **B**

> A giant metallic structure contains positive metal ions and delocalised electrons throughout the lattice.
> The mobile electrons explain electrical conduction in the solid state. The metallic bonding model also allows layers of positive ions to shift while the attraction to the electron sea is maintained, explaining why the material can be drawn into wire.